

Homework

PatientID	GeneExpression	Treatment
1	8.9	Control
2	9.4	Control
3	10.1	Control
4	11.9	Drug
5	12.5	Drug
6	13.4	Drug
7	14.1	Drug
8	15.2	Drug
9	17.6	Drug
10	18.5	Drug
11	19.0	Drug
12	20.3	Drug
13	21.2	Drug
14	22.8	Drug
15	23.7	Drug

Data creation:

- Create a dataframe with 3 columns as shown in the figure beside

Plotting with R-base:

- Graph a Bar Chart for Treatment
- Graph a Box Plot for GeneExpression on 2 groups of Treatment

Plotting with ggplot2:

- Graph a Bar Chart for number of Treatment using geom_col and geom_bar. Flip the coordinate of the axis using coord_flip
- Graph a Histogram for GeneExpression.
- Graph a Box Plot for GeneExpression on 2 groups of Treatment, each group has different color

PatientID	GeneExpression	Condition
1	8.5	Placebo
2	9.1	Placebo
3	10.2	Placebo
4	11.4	Placebo
5	12.0	Placebo
6	15.6	Low_Dose
7	16.2	Low_Dose
8	14.8	Low_Dose
9	17.1	Low_Dose
10	15.9	Low_Dose
11	16.5	Low_Dose
12	18.0	Low_Dose
13	22.1	High_Dose
14	24.5	High_Dose
15	23.8	High_Dose
16	25.0	High_Dose
17	21.9	High_Dose
18	26.4	High_Dose
19	28.1	High_Dose
20	27.5	High_Dose

Homework

Data creation:

- Create a dataframe as shown in the figure beside
- Add a 4th column called LogExpression which is the log2 value of the GeneExpression column

Plotting with R-base:

- Graph a box plot of GeneExpression categorized by Condition
- Graph a Pie Chart for Condition

Plotting with ggplot2:

- Graph a Bar Chart for number of Condition using geom_col and geom_bar
- Graph a Pie Chart for Condition
- Graph a Density plot for LogExpression, categorize by Condition.
- Graph a Box Plot for LogExpression, categorize by Condition